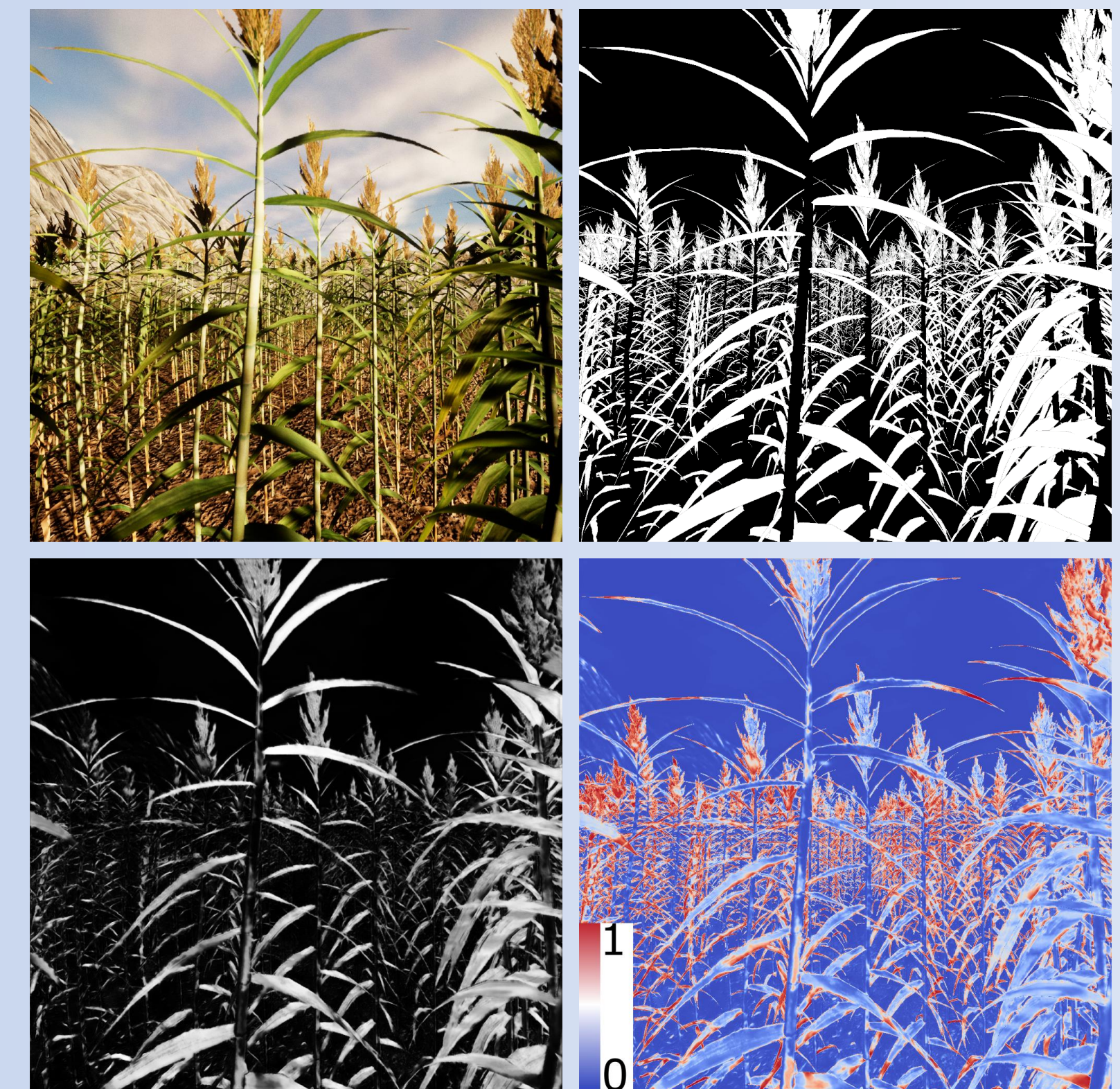


TOWARDS LARGE-SCALE RENDERING OF SIMULATED CROPS FOR SYNTHETIC GROUND TRUTH GENERATION ON MODULAR SUPERCOMPUTERS

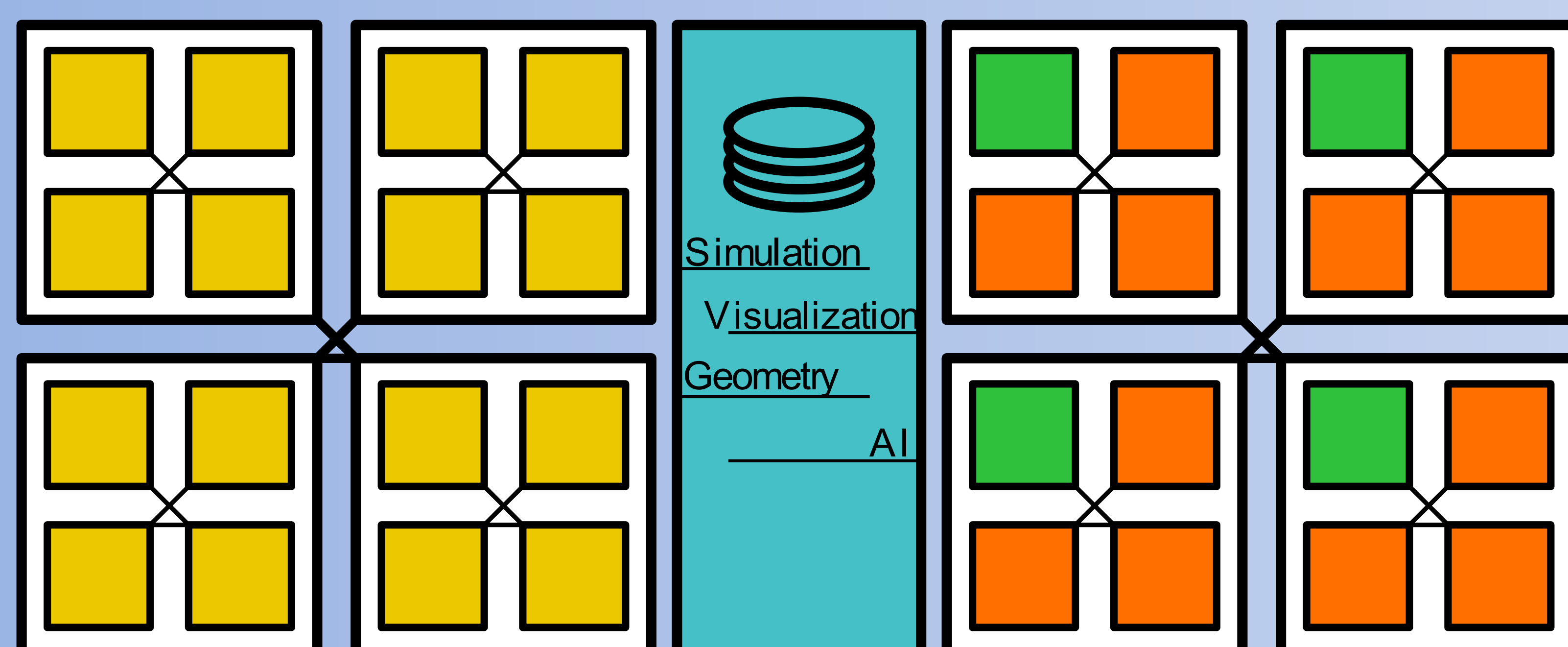
Dirk Norbert Helmrich, Jens Henrik Göbbert, Mona Giraud, Hanno Scharr, Andrea Schnepf, and Morris Riedel

VIRTUAL WORLDS FOR AI

Automated algorithms working on plant images have to overcome challenges, ranging from noisy imaging over weather conditions to “noisy” plant arrangements in the image. We are using the Unreal Engine to generate data that augments precisely these conditions, training more robust and reliable algorithms. The righthand figure shows a sample U-NET trained on virtual reed plants. We generate a segmentation map of the plant without its stem and train a U-NET on the data. The U-NET in question has learned the segmentation very well, making mistakes on the stem itself and in background regions. We believe that this network is at capacity and larger networks are needed in this case.



MODULARITY

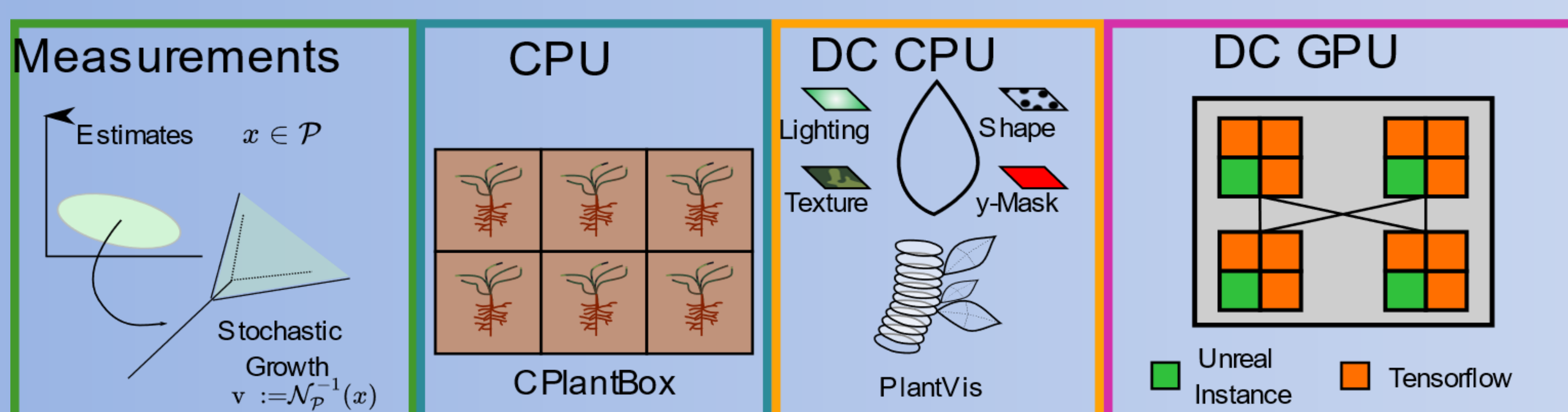


We are employing the modular supercomputer JURECA for this task. It consists of 1640 compute nodes with a total of 111520 cores. Its date-centric module consists of three types of compute nodes: 480 standard nodes with 128 cores each, 69 large-memory (1 TB) compute nodes, and 192 accelerated compute nodes with four NVIDIA A100.



The Unreal Engine runs on the GPU nodes on one graphics card. As current neural networks architectures have a hard time dealing with large resolutions, we are currently rendering 1024^2 images. Tensorflow has access to the other three GPUs. We limit node communication only to machine learning.

ARCHITECTURE



We have multiple components going into the generation of our virtual scenes. At the beginning, we are employing plant-level measurements to quantify tropism parameters of plants. This is important for the morphological correctness of generated plants. We can use CPlantBox to inform our virtual scene with plant shape and certain attributes derived from simulations. To enable a high-quality plant, multiple approaches are combined: Texture generation, mesh refinement, expedient label sets, and lighting conditions. We are currently working on an expedient connection between the Unreal Engine and our learning algorithms.

CPlantBox



Unreal
Engine



WebRTC



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Mitglied der Helmholtz-Gemeinschaft

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